

Hydraulic Analysis of the Tunduma Water Supply Transmission System

From the Momba River Intake to the Storage Tanks

Technical Report – Design Year 2035

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Abstract. This report presents a steady-state hydraulic analysis of the Tunduma Water Supply transmission system. The system abstracts water from the Momba (Saisi) River and treats it at the Momba Water Treatment Plant (20,000 m³/d net). The treated water is pumped to Ikana Ground-Level Balancing Reservoir (GBR) and conveyed by gravity to the Nkangamo Booster Pump Station (BPS). From there it is pumped to Uhuru Park GBR and four existing storage tanks in Tunduma. The pipeline geometry was taken from the topographic survey of the route. Facility data, flows and pump duties follow the Employer's Requirements (ER). Design flows are obtained from a mass balance of the tank demands. Friction losses are computed with the Hazen–Williams equation, and the hydraulic grade line is traced along 2,336 surveyed points. Pump heads are set by the most critical point of each pressure zone. Pipe pressure classes are checked to EN 805, and surge is estimated with the Joukowsky equation. The results were verified independently with the EPANET 2.2 solver. Each calculation is shown step by step, from the intake to the last storage tank.

Nomenclature

Table 1. Symbols and units.

Symbol	Description	Unit
A	internal cross-sectional area of the pipe	m ²
a	pressure wave speed	m/s
C	Hazen–Williams roughness coefficient	–
D	internal pipe diameter	m
g	gravitational acceleration (9.81)	m/s ²
H	hydraulic (piezometric) head, HGL	m a.s.l.
h_f, h_m, h_L	friction, minor and total head loss	m
h_{st}	head loss in pump-station pipework	m
L	pipe length (along the slope)	m
p	pressure head	m
P_h, P_s	hydraulic and shaft power	kW
Q	flow rate	m ³ /s
Re	Reynolds number	–
t	operating hours per day	h/d
TDH	total dynamic head of a pump	m
V	mean velocity	m/s
V_d	daily volume	m ³ /d
z	elevation (ground or pipe crown)	m a.s.l.
Δh	surge head	m
η_p, η_m	pump and motor efficiency	–
ν	kinematic viscosity of water	m ² /s
ρ	density of water (1,000)	kg/m ³

1. Introduction

The Tunduma Water Supply Project (Phase 1) supplies Tunduma town, Songwe Region, from the Momba (Saisi) River. The works comprise:

- a river intake with low-lift pumps and a grit tank with high-lift pumps;
- a raw-water main to the Momba WTP;
- a treated-water rising main to Ikana GBR;
- a gravity main to Nkangamo BPS;
- a rising main from the BPS to Uhuru Park GBR, with branches to the existing Tunduma tanks.

The objective of this study is to verify the hydraulic performance of the whole chain, from the intake to the storage tanks. It determines:

- the flows, velocities and head losses in every main;
- the hydraulic grade line and pressures along the surveyed route;
- the heads, power and energy required at each pump station;
- the pipe pressure classes;
- the storage needed to balance pumping against supply.

Data sources. The analysis uses three sources:

- the Employer's Requirements and Specifications, in particular the schematic diagram of the waterworks (Section 2.1) and Tables 7 and 8 (design criteria);

- the topographic survey of the pipeline route (2,336 alignment nodes, levels in m a.s.l.);
- a draft EPANET model derived from the route drawing (DXF). Only its geometry was used: node coordinates, ground levels and pipe lengths.

Parameters not given in the ER are stated as assumptions in Section 3 and should be confirmed during detailed design.

2. System Description

Water flows through seven hydraulic zones separated by free water surfaces: the river, the intake sump, the grit tank, the WTP clear-water tank, Ikana GBR, the Nkangamo sump and Uhuru Park GBR. Each zone is either pumped or gravity-fed. Figure 1 shows the arrangement; Table 2 lists the facilities.

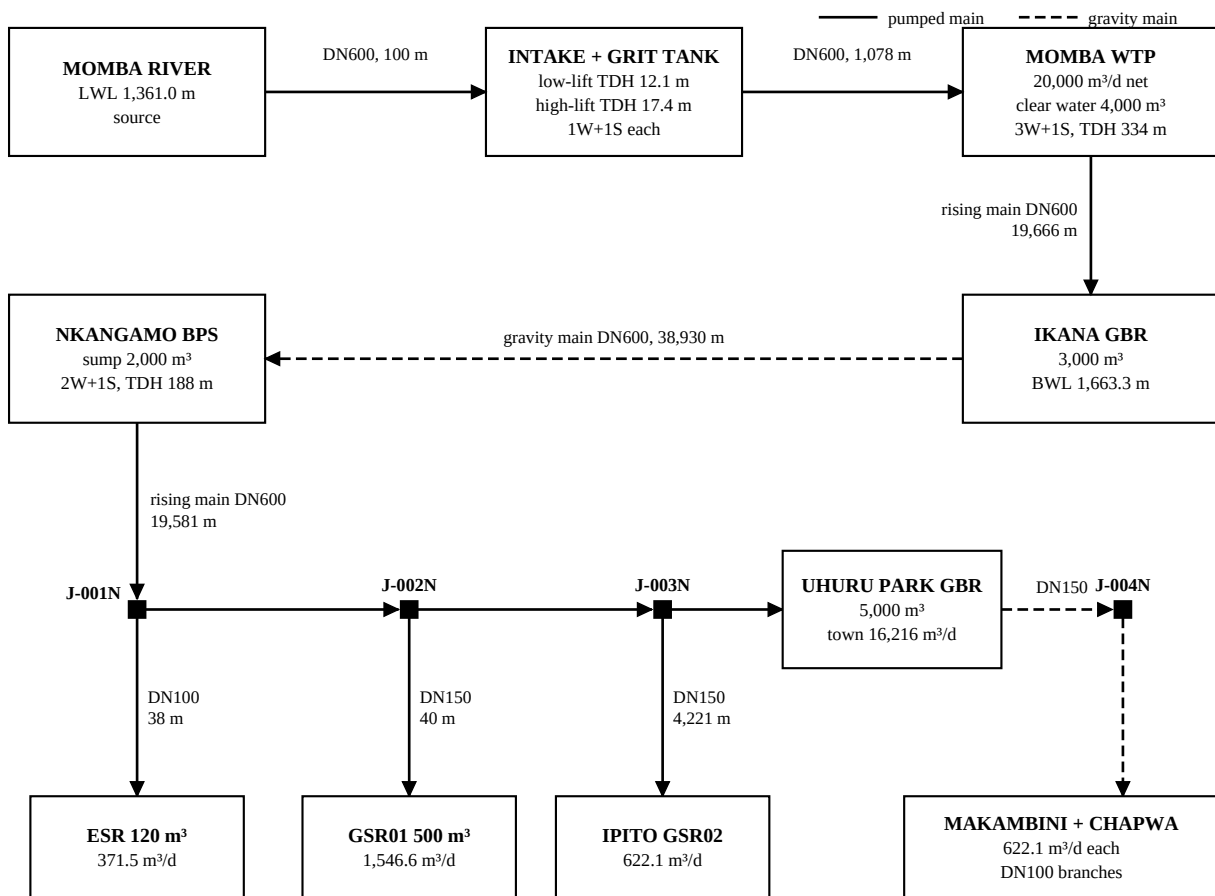


Figure 1. Schematic of the Tunduma transmission system (after ER Figure 2). Solid lines: pumped mains; dashed lines: gravity mains.

Table 2. Facilities of the system (levels in m a.s.l.). LWL/BWL = lowest water level; TWL = top water level.

Facility	Type	Ground	LWL/BWL	TWL	Volume (m ³)	Demand (m ³ /d)
MOMBA_RIVER	SOURCE	1,363.00	1,361.00	1,364.50	–	–
MOMBA_INTAKE	INTAKE	1,365.94	1,360.00	1,364.00	–	–
GRIT_TANK	SUMP	1,365.94	1,366.50	1,368.00	–	–
MOMBA_WTP	WTP	1,372.10	1,368.50	1,373.00	4,000	–
IKANA_GBR	TANK	1,663.00	1,663.30	1,668.00	3,000	–
NKANGAMO_BPS	BPS	1,555.50	1,553.00	1,557.00	2,000	–
J-001N	JUNCTION	1,698.34	–	–	–	–
J-002N	JUNCTION	1,699.25	–	–	–	–
J-003N	JUNCTION	1,699.34	–	–	–	–
UHURU_PARK_GBR	TANK	1,697.30	1,697.60	1,702.30	5,000	16,215.62
TUNDUMA_ESR	TANK	1,700.00	1,703.50	1,706.50	120	371.52
UHURU_GSR01	TANK	1,698.40	1,698.40	1,701.40	500	1,546.56
IPITO_GSR02	TANK	1,681.20	1,681.20	1,684.20	200	622.10
J-004N	JUNCTION	1,630.08	–	–	–	–
MAKAMBINI_GSR03	TANK	1,655.70	1,655.70	1,658.70	200	622.10
CHAPWA_GSR04	TANK	1,521.40	1,521.40	1,524.40	200	622.10

Table 3. Pipelines. Model length is measured along the surveyed ground slope; ER length from the schematic.

Pipe	From → To	Size	Model L (m)	ER L (m)	Flow
ABSTRACTION	MOMBA_RIVER → MOMBA_INTAKE	DN600	100	100	gravity
LL_DELIVERY	MOMBA_INTAKE → GRIT_TANK	DN600	30	–	pumped
RAW_MAIN	GRIT_TANK → MOMBA_WTP	DN600	1,078	1,177	pumped
WTP_IKANA_RM	MOMBA_WTP → IKANA_GBR	DN600	19,666	20,994	pumped
IKANA_NKANGAMO_GM	IKANA_GBR → NKANGAMO_BPS	DN600	38,930	38,504	gravity
BPS_RM_1	NKANGAMO_BPS → J-001N	DN600	19,581	21,673	pumped
BPS_RM_2	J-001N → J-002N	DN600	40	–	pumped
BPS_RM_3	J-002N → J-003N	DN600	100	–	pumped
UHURU_INLET	J-003N → UHURU_PARK_GBR	DN600	50	–	pumped
ESR_BRANCH	J-001N → TUNDUMA_ESR	DN100	38	38	pumped
GSR01_BRANCH	J-002N → UHURU_GSR01	DN150	40	40	pumped
IPITO_BRANCH	J-003N → IPITO_GSR02	DN150	4,221	4,285	pumped
UHURU_J004N	UHURU_PARK_GBR → J-004N	DN150	1,563	1,555	gravity
MAKAMBINI_BRANCH	J-004N → MAKAMBINI_GSR03	DN100	494	494	gravity
CHAPWA_BRANCH	J-004N → CHAPWA_GSR04	DN100	2,790	2,664	gravity

3. Design Criteria and Assumptions

Table 4. Design criteria adopted.

Parameter	Value	Source
Net WTP output (peak day)	20,000 m ³ /d	ER Table 8
Losses intake → WTP + within WTP	2 % + 5 % = 7 %	ER Table 8
Pump operating time	22 h/d	ER Table 8
Gravity main operating time	24 h/d	continuous supply
Friction formula	Hazen–Williams	–
Hazen–Williams C (cement-lined steel, aged)	120	assumed
Minor losses	10 % of friction	assumed
Losses inside pump stations	3.0 m	assumed
Minimum pressure at junctions / delivery	5 m	ER Table 7
Minimum pressure at the pipe crown along mains	2 m	assumed
Depth of pipe crown below ground	1.2 m	assumed
Residual head above tank inlet level	1.0 m	assumed
Maximum velocity (transmission)	3.0 m/s	ER Table 7
Pressure wave speed (steel)	1000 m/s	assumed
Pipe pressure classes	PN10, PN16, PN25, PN30, PN40, PN64	ER schematic
Water temperature	20 °C ($\nu = 1.017\text{e-}06 \text{ m}^2/\text{s}$)	assumed

Assumed levels. The ER gives ground elevations and capacities but not the operating water levels. The following were assumed:

- river low water level 1,361.0 m, taken from the survey points of the river bed (1,360.7–1,364.5 m);
- intake sump LWL 1,360.0 m;
- grit tank 1,366.5–1,368.0 m;
- WTP inlet (cascade aerator) 1,378.0 m;
- clear-water tank 1,368.5–1,373.0 m;
- new GBRs 5 m deep above the ER elevation;
- existing tanks 3 m deep, with the ESR floor at its ER elevation of 1,703.5 m.

These assumptions influence the pump heads and should be replaced by the final civil designs.

4. Methodology

4.1 Continuity and design flows

Flows are derived backwards from the demands, i.e. the outflow of each storage tank to its distribution zone. The daily volume entering a node equals its own demand plus the volumes passed on to the downstream nodes, divided by $(1 - \text{losses})$:

$$V_{d,in} = (V_{d,node} + \Sigma V_{d,out}) / (1 - \lambda) \quad (1)$$

where λ is the fractional loss at the node (7 % at the WTP). The instantaneous design flow of a main is the daily volume divided by its operating time t :

$$Q = V_d / (3600 t) \quad (2)$$

4.2 Velocity and flow regime

$$A = \pi D^2 / 4, \quad V = Q / A, \quad Re = V D / \nu \quad (3)$$

4.3 Friction and minor losses

Friction losses are computed with the Hazen–Williams equation in SI units, valid for fully turbulent flow of water:

$$h_f = 10.67 L Q^{1.852} / (C^{1.852} D^{4.8704}) \quad (4)$$

Minor losses at bends, valves and fittings are taken as a fraction k_m of the friction loss:

$$h_m = k_m h_f, \quad h_L = h_f + h_m \quad (5)$$

Pipe lengths are measured along the ground slope between successive survey points: $L = \sum (\Delta L_{plan}^2 + \Delta z^2)^{1/2}$.

4.4 Energy equation and hydraulic grade line

Velocity heads are small ($V^2/2g < 0.07$ m for $V < 1.2$ m/s), so energy and hydraulic grade lines are taken as equal. Along a pipe of uniform flow the HGL falls linearly with chainage x :

$$H(x) = H_0 - h_L x / L, \quad p(x) = H(x) - z_c(x) \quad (6)$$

Here z_c is the pipe crown level (ground level minus cover). A zone starts at a free water surface and ends at the next one. The head required at the start of a zone is the largest of two values, taken over every downstream point:

- (i) the inlet level of each receiving tank plus a residual head, plus the losses on the way;
- (ii) the level needed to keep the minimum pressure p_{min} at every survey point of every pipe.

$$H_{req} = \max [z_{inlet} + h_{res} + \sum h_L; z_c(x) + p_{min} + h_L x/L + \sum h_{L,upstream}] \quad (7)$$

In a gravity zone, the available head is the bottom water level (BWL) of the supplying tank, and the surplus $H_{avail} - H_{req}$ must be dissipated by a flow-control or pressure-reducing valve. In a pumped zone, the pumps must supply H_{req} .

4.5 Pump head, power and energy

$$TDH = H_{req} - z_{LWL,suction} + h_{st} \quad (8)$$

$$P_h = \rho g Q_{pump} TDH / 1000, \quad P_s = P_h / \eta_p, \quad P_{motor} \geq 1.15 P_s \quad (9)$$

$$E = n_{duty} P_s t / \eta_m \text{ (kWh/d)}, \quad e = E / V_d \text{ (kWh/m}^3\text{)} \quad (10)$$

4.6 Pressure class and surge

The working pressure at each point is the larger of the steady-state pressure and the static pressure with the flow stopped. For a gravity zone, the static level is the TWL of the supplying tank; for a pumped zone, it is the highest TWL at the end of the zone. Following EN 805, the allowable operating pressure PFA (= PN, bar) must cover the working pressure. The allowable maximum pressure PMA = 1.2 PFA must also cover the working pressure plus surge. The surge caused by sudden pump trip or valve closure is estimated with the Joukowsky equation:

$$\Delta h = a \Delta V / g, \quad p_{work} \cdot 0.0981 \leq PFA, \quad (p_{work} + \Delta h) \cdot 0.0981 \leq 1.2 PFA \quad (11)$$

4.7 Storage

A tank filled for t hours per day but drawn at a uniform rate over 24 hours must store the volume delivered during the pumping-off period:

$$V_{\text{bal}} = V_{\text{d,in}} (1 - t / 24) \quad (12)$$

4.8 Computation

The calculations were implemented in Python (water_hydraulics.py). The same network was also written as an EPANET 2.2 input file and solved with EPANET through the WNTR library, as an independent check (Section 9).

5. Design Flows

Table 5 gives the demand of each storage tank (ER schematic). Working back up the system with Equation (1):

Table 5. Demands supplied by the storage tanks (ER schematic, design year 2035).

Tank / zone	Demand (m ³ /d)
UHURU_PARK_GBR	16,215.62
UHURU_GSR01	1,546.56
TUNDUMA_ESR	371.52
IPITO_GSR02	622.10
MAKAMBINI_GSR03	622.10
CHAPWA_GSR04	622.10
Total delivered	20,000.00

Treated water leaving the WTP: $V_d = 20,000.00 \text{ m}^3/\text{d}$

Raw water at the intake: $V_d = 20,000 / (1 - 0.07) = \mathbf{21,505.4 \text{ m}^3/\text{d}}$ (ER intake capacity 21,506 m³/d)

Flow in pumped mains ($t = 22 \text{ h}$): $Q_{\text{raw}} = 21,505.4 / (22 \times 3600) = 0.2715 \text{ m}^3/\text{s}$; $Q_{\text{treated}} = 20,000 / 79,200 = 0.2525 \text{ m}^3/\text{s}$

Flow in the gravity main ($t = 24 \text{ h}$): $Q = 20,000 / 86,400 = 0.2315 \text{ m}^3/\text{s}$

Branch J-004N (gravity from Uhuru Park, 24 h): $V_d = 622.1 + 622.1 = 1,244.2 \text{ m}^3/\text{d} \rightarrow Q = 14.40 \text{ l/s}$

6. Step-by-Step Hydraulic Calculations

The calculations follow the water from the river to the last tank. For each pipe, the design flow, velocity and head loss are computed with Equations (2)–(5). For each zone, the head required is then found with Equation (7). Finally the pump head and power are computed with Equations (8)–(10), or the gravity surplus is determined.

6.1 Zone 1 – River to intake sump (gravity abstraction)

Water enters the intake sump by gravity through the abstraction pipe. The river low water level must exceed the sump LWL plus the losses in the abstraction pipe.

Pipe ABSTRACTION: MOMBA_RIVER → MOMBA_INTAKE (DN600, $L = 100 \text{ m}$, $C = 120$)

Design flow: $Q = V_d / (t \times 3600) = 21,505.38 / (22 \times 3600) = \mathbf{0.2715 \text{ m}^3/\text{s}}$ (271.5 l/s)

Area: $A = \pi D^2/4 = \pi \times 0.600^2 / 4 = 0.2827 \text{ m}^2$; velocity $V = Q/A = 0.2715 / 0.2827 = \mathbf{0.960 \text{ m/s}}$

Reynolds number: $Re = VD/\nu = 0.960 \times 0.600 / 1.017\text{e-}06 = 566,692$ (turbulent, $Re > 4,000$)

Friction: $h_f = 10.67 \text{ L } Q^{1.852} / (C^{1.852} D^{4.8704}) = 10.67 \times 100.0 \times 0.08942 / (7,090.0 \times 0.08308) = \mathbf{0.16 \text{ m}}$

Minor losses: $h_m = k_m h_f = 0.10 \times 0.16 = 0.02 \text{ m}$; total $h_L = 0.16 + 0.02 = \mathbf{0.18 \text{ m}}$ (gradient 1.78 m/km)

HGL: $H_{\text{start}} = 1361.00 \text{ m}$, $H_{\text{end}} = 1361.00 - 0.18 = 1360.82 \text{ m}$; pressure along the pipe up to 2.0 m
Pressure class (working pressure): PN10 (0-100 m); ER: PN10 100 m

Available: river LWL = 1361.00 m; required: sump LWL + $h_L = 1360.00 + 0.18 = 1360.18 \text{ m}$; surplus = **0.82 m** → the sump fills at the design flow.

6.2 Zone 2 – Intake low-lift pumps to grit tank

Pipe LL_DELIVERY: MOMBA_INTAKE → GRIT_TANK (DN600, $L = 30 \text{ m}$, $C = 120$)

Design flow: $Q = V_d / (t \times 3600) = 21,505.38 / (22 \times 3600) = \mathbf{0.2715 \text{ m}^3/\text{s}}$ (271.5 l/s)

Area: $A = \pi D^2/4 = \pi \times 0.600^2 / 4 = 0.2827 \text{ m}^2$; velocity $V = Q/A = 0.2715 / 0.2827 = \mathbf{0.960 \text{ m/s}}$

Reynolds number: $Re = VD/\nu = 0.960 \times 0.600 / 1.017\text{e-}06 = 566,692$ (turbulent, $Re > 4,000$)

Friction: $h_f = 10.67 L Q^{1.852} / (C^{1.852} D^{4.8704}) = 10.67 \times 30.0 \times 0.08942 / (7,090.0 \times 0.08308) = \mathbf{0.05 \text{ m}}$

Minor losses: $h_m = k_m h_f = 0.10 \times 0.05 = 0.00 \text{ m}$; total $h_L = 0.05 + 0.00 = \mathbf{0.05 \text{ m}}$ (gradient 1.78 m/km)

HGL: $H_{\text{start}} = 1369.05 \text{ m}$, $H_{\text{end}} = 1369.05 - 0.05 = 1369.00 \text{ m}$; pressure along the pipe up to 4.3 m
Pressure class (working pressure): PN10 (0-30 m)

Pump station MOMBA_INTAKE

Controlling point: inlet of GRIT_TANK: $H = \text{inlet level} + \text{residual} = 1368.00 + 1.00 = 1369.00 \text{ m}$
+ head loss in LL_DELIVERY: 0.05 m → 1369.05 m

Head required at the start of the zone: $H_{\text{req}} = \mathbf{1369.05 \text{ m}}$

TDH = $H_{\text{req}} - z_{\text{LWL}} + h_{\text{st}} = 1369.05 - 1360.00 + 3.0 = \mathbf{12.1 \text{ m}}$ (ER duty head 10 m)

Flow per pump: $Q_{\text{pump}} = 0.2715 / 1 = 0.2715 \text{ m}^3/\text{s} = 978 \text{ m}^3/\text{h}$ (ER 978 m³/h)

$P_h = 1000 \times 9.81 \times 0.2715 \times 12.1 / 1000 = 32.1 \text{ kW}$; $P_s = 32.1 / 0.75 = \mathbf{42.8 \text{ kW}}$

Motor: $1.15 \times 42.8 = 49.2 \text{ kW}$ → standard motor **55 kW** per pump (1 duty + 1 standby)

Energy: $E = 1 \times 42.8 \times 22 / 0.94 = 1,002 \text{ kWh/d}$; specific energy $e = 1,002 / 21,505 = 0.047 \text{ kWh/m}^3$

6.3 Zone 3 – Grit tank high-lift pumps and raw-water main to the WTP

Pipe RAW_MAIN: GRIT_TANK → MOMBA_WTP (DN600, $L = 1,078 \text{ m}$, $C = 120$)

The raw-water main follows the surveyed route for 1,078 m and discharges into the WTP inlet (cascade aerator).

Design flow: $Q = V_d / (t \times 3600) = 21,505.38 / (22 \times 3600) = \mathbf{0.2715 \text{ m}^3/\text{s}}$ (271.5 l/s)

Area: $A = \pi D^2/4 = \pi \times 0.600^2 / 4 = 0.2827 \text{ m}^2$; velocity $V = Q/A = 0.2715 / 0.2827 = \mathbf{0.960 \text{ m/s}}$

Reynolds number: $Re = VD/\nu = 0.960 \times 0.600 / 1.017\text{e-}06 = 566,692$ (turbulent, $Re > 4,000$)

Friction: $h_f = 10.67 L Q^{1.852} / (C^{1.852} D^{4.8704}) = 10.67 \times 1,078.5 \times 0.08942 / (7,090.0 \times 0.08308) = \mathbf{1.75 \text{ m}}$

Minor losses: $h_m = k_m h_f = 0.10 \times 1.75 = 0.17 \text{ m}$; total $h_L = 1.75 + 0.17 = \mathbf{1.92 \text{ m}}$ (gradient 1.78 m/km)

HGL: $H_{\text{start}} = 1380.92 \text{ m}$, $H_{\text{end}} = 1380.92 - 1.92 = 1379.00 \text{ m}$; pressure along the pipe 8.3 to 16.2 m

Surge (Joukowsky): $\Delta h = aV/g = 1000 \times 0.960 / 9.81 = \mathbf{97.9 \text{ m}}$; minimum transient pressure $\approx p_{\text{min}} - \Delta h = 8.3 - 97.9 = -89.6 \text{ m}$

Pressure class (working pressure): PN10 (0-1,078 m); ER: PN10 1,177 m

Pump station GRIT_TANK

Controlling point: inlet of MOMBA_WTP: $H = \text{inlet level} + \text{residual} = 1378.00 + 1.00 = 1379.00 \text{ m}$
+ head loss in RAW_MAIN: 1.92 m → 1380.92 m

Head required at the start of the zone: $H_{\text{req}} = \mathbf{1380.92 \text{ m}}$

$TDH = H_{req} - z_{LWL} + h_{st} = 1380.92 - 1366.50 + 3.0 = \mathbf{17.4\ m}$ (ER duty head 25 m)
 Flow per pump: $Q_{pump} = 0.2715 / 1 = 0.2715\ m^3/s = 978\ m^3/h$ (ER 978 m³/h)
 $P_h = 1000 \times 9.81 \times 0.2715 \times 17.4 / 1000 = 46.4\ kW$; $P_s = 46.4 / 0.75 = \mathbf{61.9\ kW}$
 Motor: $1.15 \times 61.9 = 71.2\ kW \rightarrow$ standard motor **75 kW** per pump (1 duty + 1 standby)
 Energy: $E = 1 \times 61.9 \times 22 / 0.94 = 1,448\ kWh/d$; specific energy $e = 1,448 / 21,505 = 0.067\ kWh/m^3$

6.4 Zone 4 – WTP clear-water pumps and rising main to Ikana GBR

The WTP delivers 20,000 m³/d. The clear-water tank (4,000 m³) supplies the treated-water rising main, which climbs from 1,368.5 m (CWT LWL) to Ikana GBR (TWL 1,668.0 m).

Pipe WTP_IKANA_RM: MOMBA_WTP → IKANA_GBR (DN600, L = 19,666 m, C = 120)

Design flow: $Q = V_d / (t \times 3600) = 20,000.00 / (22 \times 3600) = \mathbf{0.2525\ m^3/s}$ (252.5 l/s)
 Area: $A = \pi D^2/4 = \pi \times 0.600^2 / 4 = 0.2827\ m^2$; velocity $V = Q/A = 0.2525 / 0.2827 = \mathbf{0.893\ m/s}$
 Reynolds number: $Re = VD/\nu = 0.893 \times 0.600 / 1.017e-06 = 527,024$ (turbulent, $Re > 4,000$)
 Friction: $h_f = 10.67 L Q^{1.852} / (C^{1.852} D^{4.8704}) = 10.67 \times 19,665.6 \times 0.07818 / (7,090.0 \times 0.08308) = \mathbf{27.85\ m}$
 Minor losses: $h_m = k_m h_f = 0.10 \times 27.85 = 2.78\ m$; total $h_L = 27.85 + 2.78 = \mathbf{30.63\ m}$ (gradient 1.56 m/km)
 HGL: $H_{start} = 1699.63\ m$, $H_{end} = 1699.63 - 30.63 = 1669.00\ m$; pressure along the pipe 8.9 to 328.6 m
 Surge (Joukowsky): $\Delta h = aV/g = 1000 \times 0.893 / 9.81 = \mathbf{91.0\ m}$; minimum transient pressure $\approx p_{min} - \Delta h = 8.9 - 91.0 = -82.2\ m$
 Pressure class (working pressure): PN40 (0-1,450 m), PN30 (1,450-12,227 m), PN25 (12,227-18,973 m), PN16 (18,973-19,408 m), PN10 (19,408-19,666 m); ER: PN10 305 m, PN16 119 m, PN25 4,705 m, PN30 10,798 m, PN40 5,067 m

Pump station MOMBA_WTP

Controlling point: inlet of IKANA_GBR: $H = \text{inlet level} + \text{residual} = 1668.00 + 1.00 = 1669.00\ m$
 + head loss in WTP_IKANA_RM: 30.63 m → 1699.63 m
 Head required at the start of the zone: $H_{req} = \mathbf{1699.63\ m}$
 $TDH = H_{req} - z_{LWL} + h_{st} = 1699.63 - 1368.50 + 3.0 = \mathbf{334.1\ m}$ (ER duty head 330 m)
 Flow per pump: $Q_{pump} = 0.2525 / 3 = 0.0842\ m^3/s = 303\ m^3/h$ (ER 304 m³/h)
 $P_h = 1000 \times 9.81 \times 0.0842 \times 334.1 / 1000 = 275.9\ kW$; $P_s = 275.9 / 0.78 = \mathbf{353.7\ kW}$
 Motor: $1.15 \times 353.7 = 406.8\ kW \rightarrow$ standard motor **450 kW** per pump (3 duty + 1 standby)
 Energy: $E = 3 \times 353.7 \times 22 / 0.95 = 24,575\ kWh/d$; specific energy $e = 24,575 / 20,000 = 1.229\ kWh/m^3$

6.5 Zone 5 – Ikana GBR to Nkangamo BPS (gravity main)

Pipe IKANA_NKANGAMO_GM: IKANA_GBR → NKANGAMO_BPS (DN600, L = 38,930 m, C = 120)

Design flow: $Q = V_d / (t \times 3600) = 20,000.00 / (24 \times 3600) = \mathbf{0.2315\ m^3/s}$ (231.5 l/s)
 Area: $A = \pi D^2/4 = \pi \times 0.600^2 / 4 = 0.2827\ m^2$; velocity $V = Q/A = 0.2315 / 0.2827 = \mathbf{0.819\ m/s}$
 Reynolds number: $Re = VD/\nu = 0.819 \times 0.600 / 1.017e-06 = 483,105$ (turbulent, $Re > 4,000$)
 Friction: $h_f = 10.67 L Q^{1.852} / (C^{1.852} D^{4.8704}) = 10.67 \times 38,930.2 \times 0.06654 / (7,090.0 \times 0.08308) = \mathbf{46.92\ m}$
 Minor losses: $h_m = k_m h_f = 0.10 \times 46.92 = 4.69\ m$; total $h_L = 46.92 + 4.69 = \mathbf{51.62\ m}$ (gradient 1.33 m/km)

HGL: $H_{\text{start}} = 1663.30 \text{ m}$, $H_{\text{end}} = 1663.30 - 51.62 = 1611.68 \text{ m}$; pressure along the pipe 65.6 to 205.9 m
 Surge (Joukowsky): $\Delta h = aV/g = 1000 \times 0.819 / 9.81 = \mathbf{83.5 \text{ m}}$; minimum transient pressure $\approx p_{\text{min}} - \Delta h$
 $= 65.6 - 83.5 = -17.8 \text{ m}$

Pressure class (working pressure): PN10 (0-170 m), PN16 (170-381 m), PN25 (381-7,257 m), PN16 (7,257-9,906 m), PN25 (9,906-10,006 m), PN16 (10,006-10,706 m), PN25 (10,706-12,707 m), PN16 (12,707-13,307 m), PN25 (13,307-14,358 m), PN16 (14,358-19,259 m), PN10 (19,259-19,659 m), PN16 (19,659-24,061 m), PN25 (24,061-24,611 m), PN16 (24,611-25,611 m), PN25 (25,611-34,667 m), PN16 (34,667-35,217 m), PN25 (35,217-35,367 m), PN16 (35,367-38,930 m); ER: PN10 2,877 m, PN16 13,247 m, PN25 22,380 m

Controlling point: inlet of NKANGAMO_BPS: $H = \text{inlet level} + \text{residual} = 1557.50 + 1.00 = 1558.50 \text{ m}$
 $+ \text{head loss in IKANA_NKANGAMO_GM: } 51.62 \text{ m} \rightarrow 1610.12 \text{ m}$

Head required at the start of the zone: $H_{\text{req}} = \mathbf{1610.12 \text{ m}}$

Available head: Ikana BWL = 1663.30 m; surplus = $1663.30 - 1610.12 = \mathbf{53.2 \text{ m}}$

The surplus is dissipated by a flow-control valve at the Nkangamo sump inlet, set to the design flow of 231.5 l/s.

6.6 Zone 6 – Nkangamo BPS to Uhuru Park GBR and the existing tanks

The BPS rising main supplies three offtakes before Uhuru Park GBR: J-001N (existing ESR), J-002N (existing Uhuru GSR 01) and J-003N (Ipito GSR 02). Every branch must be satisfied. The branch that needs the highest head at the pumps governs, and flow-control valves throttle the other branches.

Pipe BPS_RM_1: NKANGAMO_BPS $\rightarrow$ J-001N (DN600, $L = 19,581 \text{ m}$, $C = 120$)

Design flow: $Q = V_d / (t \times 3600) = 20,000.00 / (22 \times 3600) = \mathbf{0.2525 \text{ m}^3/\text{s}}$ (252.5 l/s)

Area: $A = \pi D^2/4 = \pi \times 0.600^2 / 4 = 0.2827 \text{ m}^2$; velocity $V = Q/A = 0.2525 / 0.2827 = \mathbf{0.893 \text{ m/s}}$

Reynolds number: $Re = VD/\nu = 0.893 \times 0.600 / 1.017\text{e-}06 = 527,024$ (turbulent, $Re > 4,000$)

Friction: $h_f = 10.67 L Q^{1.852} / (C^{1.852} D^{4.8704}) = 10.67 \times 19,581.4 \times 0.07818 / (7,090.0 \times 0.08308) = \mathbf{27.73 \text{ m}}$

Minor losses: $h_m = k_m h_f = 0.10 \times 27.73 = 2.77 \text{ m}$; total $h_L = 27.73 + 2.77 = \mathbf{30.50 \text{ m}}$ (gradient 1.56 m/km)

HGL: $H_{\text{start}} = 1738.23 \text{ m}$, $H_{\text{end}} = 1738.23 - 30.50 = 1707.73 \text{ m}$; pressure along the pipe 6.7 to 183.9 m
 Surge (Joukowsky): $\Delta h = aV/g = 1000 \times 0.893 / 9.81 = \mathbf{91.0 \text{ m}}$; minimum transient pressure $\approx p_{\text{min}} - \Delta h$
 $= 6.7 - 91.0 = -84.4 \text{ m}$

Pressure class (working pressure): PN25 (0-613 m), PN16 (613-1,513 m), PN25 (1,513-2,553 m), PN16 (2,553-5,495 m), PN25 (5,495-5,776 m), PN16 (5,776-7,558 m), PN10 (7,558-19,581 m); ER: PN10 10,310 m, PN16 7,177 m, PN25 4,185 m

Pipe BPS_RM_2: J-001N $\rightarrow$ J-002N (DN600, $L = 40 \text{ m}$, $C = 120$)

Design flow: $Q = V_d / (t \times 3600) = 19,628.48 / (22 \times 3600) = \mathbf{0.2478 \text{ m}^3/\text{s}}$ (247.8 l/s)

Area: $A = \pi D^2/4 = \pi \times 0.600^2 / 4 = 0.2827 \text{ m}^2$; velocity $V = Q/A = 0.2478 / 0.2827 = \mathbf{0.877 \text{ m/s}}$

Reynolds number: $Re = VD/\nu = 0.877 \times 0.600 / 1.017\text{e-}06 = 517,234$ (turbulent, $Re > 4,000$)

Friction: $h_f = 10.67 L Q^{1.852} / (C^{1.852} D^{4.8704}) = 10.67 \times 40.0 \times 0.07551 / (7,090.0 \times 0.08308) = \mathbf{0.05 \text{ m}}$

Minor losses: $h_m = k_m h_f = 0.10 \times 0.05 = 0.01 \text{ m}$; total $h_L = 0.05 + 0.01 = \mathbf{0.06 \text{ m}}$ (gradient 1.50 m/km)

HGL: $H_{\text{start}} = 1707.73 \text{ m}$, $H_{\text{end}} = 1707.73 - 0.06 = 1707.67 \text{ m}$; pressure along the pipe 9.6 to 10.6 m

Pressure class (working pressure): PN10 (0-40 m)

Pipe BPS_RM_3: J-002N → J-003N (DN600, L = 100 m, C = 120)

Design flow: $Q = V_d / (t \times 3600) = 18,081.92 / (22 \times 3600) = \mathbf{0.2283 \text{ m}^3/\text{s}}$ (228.3 l/s)

Area: $A = \pi D^2/4 = \pi \times 0.600^2 / 4 = 0.2827 \text{ m}^2$; velocity $V = Q/A = 0.2283 / 0.2827 = \mathbf{0.807 \text{ m/s}}$

Reynolds number: $Re = VD/\nu = 0.807 \times 0.600 / 1.017\text{e-}06 = 476,480$ (turbulent, $Re > 4,000$)

Friction: $h_f = 10.67 L Q^{1.852} / (C^{1.852} D^{4.8704}) = 10.67 \times 100.2 \times 0.06486 / (7,090.0 \times 0.08308) = \mathbf{0.12 \text{ m}}$

Minor losses: $h_m = k_m h_f = 0.10 \times 0.12 = 0.01 \text{ m}$; total $h_L = 0.12 + 0.01 = \mathbf{0.13 \text{ m}}$ (gradient 1.29 m/km)

HGL: $H_{\text{start}} = 1707.67 \text{ m}$, $H_{\text{end}} = 1707.67 - 0.13 = 1707.54 \text{ m}$; pressure along the pipe 9.2 to 9.6 m

Pressure class (working pressure): PN10 (0-100 m)

Pipe UHURU_INLET: J-003N → UHURU_PARK_GBR (DN600, L = 50 m, C = 120)

Design flow: $Q = V_d / (t \times 3600) = 17,459.82 / (22 \times 3600) = \mathbf{0.2205 \text{ m}^3/\text{s}}$ (220.5 l/s)

Area: $A = \pi D^2/4 = \pi \times 0.600^2 / 4 = 0.2827 \text{ m}^2$; velocity $V = Q/A = 0.2205 / 0.2827 = \mathbf{0.780 \text{ m/s}}$

Reynolds number: $Re = VD/\nu = 0.780 \times 0.600 / 1.017\text{e-}06 = 460,087$ (turbulent, $Re > 4,000$)

Friction: $h_f = 10.67 L Q^{1.852} / (C^{1.852} D^{4.8704}) = 10.67 \times 50.0 \times 0.06079 / (7,090.0 \times 0.08308) = \mathbf{0.06 \text{ m}}$

Minor losses: $h_m = k_m h_f = 0.10 \times 0.06 = 0.01 \text{ m}$; total $h_L = 0.06 + 0.01 = \mathbf{0.06 \text{ m}}$ (gradient 1.21 m/km)

HGL: $H_{\text{start}} = 1707.54 \text{ m}$, $H_{\text{end}} = 1707.54 - 0.06 = 1707.48 \text{ m}$; pressure along the pipe 9.4 to 11.4 m

Pressure class (working pressure): PN10 (0-50 m)

Pipe ESR_BRANCH: J-001N → TUNDUMA_ESR (DN100, L = 38 m, C = 120)

Design flow: $Q = V_d / (t \times 3600) = 371.52 / (22 \times 3600) = \mathbf{0.0047 \text{ m}^3/\text{s}}$ (4.7 l/s)

Area: $A = \pi D^2/4 = \pi \times 0.100^2 / 4 = 0.0079 \text{ m}^2$; velocity $V = Q/A = 0.0047 / 0.0079 = \mathbf{0.597 \text{ m/s}}$

Reynolds number: $Re = VD/\nu = 0.597 \times 0.100 / 1.017\text{e-}06 = 58,740$ (turbulent, $Re > 4,000$)

Friction: $h_f = 10.67 L Q^{1.852} / (C^{1.852} D^{4.8704}) = 10.67 \times 38.0 \times 0.00005 / (7,090.0 \times 0.00001) = \mathbf{0.21 \text{ m}}$

Minor losses: $h_m = k_m h_f = 0.10 \times 0.21 = 0.02 \text{ m}$; total $h_L = 0.21 + 0.02 = \mathbf{0.23 \text{ m}}$ (gradient 5.98 m/km)

HGL: $H_{\text{start}} = 1707.73 \text{ m}$, $H_{\text{end}} = 1707.73 - 0.23 = 1707.50 \text{ m}$; pressure along the pipe 10.6 to 10.6 m

Pressure class (working pressure): PN10 (0-38 m); ER: PN10 38 m

Pipe GSR01_BRANCH: J-002N → UHURU_GSR01 (DN150, L = 40 m, C = 120)

Design flow: $Q = V_d / (t \times 3600) = 1,546.56 / (22 \times 3600) = \mathbf{0.0195 \text{ m}^3/\text{s}}$ (19.5 l/s)

Area: $A = \pi D^2/4 = \pi \times 0.150^2 / 4 = 0.0177 \text{ m}^2$; velocity $V = Q/A = 0.0195 / 0.0177 = \mathbf{1.105 \text{ m/s}}$

Reynolds number: $Re = VD/\nu = 1.105 \times 0.150 / 1.017\text{e-}06 = 163,015$ (turbulent, $Re > 4,000$)

Friction: $h_f = 10.67 L Q^{1.852} / (C^{1.852} D^{4.8704}) = 10.67 \times 40.0 \times 0.00068 / (7,090.0 \times 0.00010) = \mathbf{0.42 \text{ m}}$

Minor losses: $h_m = k_m h_f = 0.10 \times 0.42 = 0.04 \text{ m}$; total $h_L = 0.42 + 0.04 = \mathbf{0.47 \text{ m}}$ (gradient 11.64 m/km)

HGL: $H_{\text{start}} = 1707.67 \text{ m}$, $H_{\text{end}} = 1707.67 - 0.47 = 1707.20 \text{ m}$; pressure along the pipe 9.6 to 10.0 m

Pressure class (working pressure): PN10 (0-40 m); ER: PN16 40 m

Pipe IPITO_BRANCH: J-003N → IPITO_GSR02 (DN150, L = 4,221 m, C = 120)

Design flow: $Q = V_d / (t \times 3600) = 622.10 / (22 \times 3600) = \mathbf{0.0079 \text{ m}^3/\text{s}}$ (7.9 l/s)

Area: $A = \pi D^2/4 = \pi \times 0.150^2 / 4 = 0.0177 \text{ m}^2$; velocity $V = Q/A = 0.0079 / 0.0177 = \mathbf{0.444 \text{ m/s}}$

Reynolds number: $Re = VD/\nu = 0.444 \times 0.150 / 1.017\text{e-}06 = 65,572$ (turbulent, $Re > 4,000$)

Friction: $h_f = 10.67 L Q^{1.852} / (C^{1.852} D^{4.8704}) = 10.67 \times 4,220.8 \times 0.00013 / (7,090.0 \times 0.00010) = \mathbf{8.27 \text{ m}}$

Minor losses: $h_m = k_m h_f = 0.10 \times 8.27 = 0.83 \text{ m}$; total $h_L = 8.27 + 0.83 = \mathbf{9.10 \text{ m}}$ (gradient 2.16 m/km)

HGL: $H_{\text{start}} = 1707.54 \text{ m}$, $H_{\text{end}} = 1707.54 - 9.10 = 1698.44 \text{ m}$; pressure along the pipe 9.0 to 97.9 m

Surge (Joukowsky): $\Delta h = aV/g = 1000 \times 0.444 / 9.81 = \mathbf{45.3 \text{ m}}$; minimum transient pressure $\approx p_{\text{min}} - \Delta h = 9.0 - 45.3 = -36.3 \text{ m}$

Pressure class (working pressure): PN10 (0-2,734 m), PN16 (2,734-2,871 m), PN10 (2,871-3,268 m),

PN16 (3,268-3,366 m), PN10 (3,366-4,221 m); ER: PN10 4,285 m

Pump station NKANGAMO_BPS

Controlling point: inlet of TUNDUMA_ESR: $H = \text{inlet level} + \text{residual} = 1706.50 + 1.00 = 1707.50 \text{ m}$

+ head loss in ESR_BRANCH: $0.23 \text{ m} \rightarrow 1707.73 \text{ m}$

+ head loss in BPS_RM_1: $30.50 \text{ m} \rightarrow 1738.23 \text{ m}$

Head required at the start of the zone: $H_{\text{req}} = 1738.23 \text{ m}$

TDH = $H_{\text{req}} - z_{\text{LWL}} + h_{\text{st}} = 1738.23 - 1553.00 + 3.0 = 188.2 \text{ m}$ (ER duty head 221 m)

Flow per pump: $Q_{\text{pump}} = 0.2525 / 2 = 0.1263 \text{ m}^3/\text{s} = 455 \text{ m}^3/\text{h}$ (ER 455 m³/h)

$P_h = 1000 \times 9.81 \times 0.1263 \times 188.2 / 1000 = 233.1 \text{ kW}$; $P_s = 233.1 / 0.78 = 298.9 \text{ kW}$

Motor: $1.15 \times 298.9 = 343.7 \text{ kW} \rightarrow$ standard motor **355 kW** per pump (2 duty + 1 standby)

Energy: $E = 2 \times 298.9 \times 22 / 0.95 = 13,844 \text{ kWh/d}$; specific energy $e = 13,844 / 20,000 = 0.692 \text{ kWh/m}^3$

Table 6. Heads arriving at the receiving tanks of Zone 6. Excess head is removed by the inlet flow-control valves.

Tank	Required (m)	Arriving HGL (m)	Excess (m)
TUNDUMA_ESR	1,707.50	1,707.50	0.00
UHURU_GSR01	1,702.40	1,707.20	4.80
IPITO_GSR02	1,685.20	1,698.44	13.24
UHURU_PARK_GBR	1,703.30	1,697.60	4.18

6.7 Zone 7 – Uhuru Park GBR to Makambini and Chapwa (gravity)

Pipe UHURU_J004N: UHURU_PARK_GBR $\rightarrow$ J-004N (DN150, $L = 1,563 \text{ m}$, $C = 120$)

Design flow: $Q = V_d / (t \times 3600) = 1,244.20 / (24 \times 3600) = 0.0144 \text{ m}^3/\text{s}$ (14.4 l/s)

Area: $A = \pi D^2/4 = \pi \times 0.150^2 / 4 = 0.0177 \text{ m}^2$; velocity $V = Q/A = 0.0144 / 0.0177 = 0.815 \text{ m/s}$

Reynolds number: $Re = VD/\nu = 0.815 \times 0.150 / 1.017\text{e-}06 = 120,216$ (turbulent, $Re > 4,000$)

Friction: $h_f = 10.67 L Q^{1.852} / (C^{1.852} D^{4.8704}) = 10.67 \times 1,563.5 \times 0.00039 / (7,090.0 \times 0.00010) = 9.41 \text{ m}$

Minor losses: $h_m = k_m h_f = 0.10 \times 9.41 = 0.94 \text{ m}$; total $h_L = 9.41 + 0.94 = 10.35 \text{ m}$ (gradient 6.62 m/km)

HGL: $H_{\text{start}} = 1697.60 \text{ m}$, $H_{\text{end}} = 1697.60 - 10.35 = 1687.25 \text{ m}$; pressure along the pipe -1.9 to 58.4 m

Surge (Joukowsky): $\Delta h = aV/g = 1000 \times 0.815 / 9.81 = 83.1 \text{ m}$; minimum transient pressure $\approx p_{\text{min}} - \Delta h = -1.9 - 83.1 = -85.0 \text{ m}$

Pressure class (working pressure): PN10 (0-1,563 m); ER: PN10 1,555 m

Pipe MAKAMBINI_BRANCH: J-004N $\rightarrow$ MAKAMBINI_GSR03 (DN100, $L = 494 \text{ m}$, $C = 120$)

Design flow: $Q = V_d / (t \times 3600) = 622.10 / (24 \times 3600) = 0.0072 \text{ m}^3/\text{s}$ (7.2 l/s)

Area: $A = \pi D^2/4 = \pi \times 0.100^2 / 4 = 0.0079 \text{ m}^2$; velocity $V = Q/A = 0.0072 / 0.0079 = 0.917 \text{ m/s}$

Reynolds number: $Re = VD/\nu = 0.917 \times 0.100 / 1.017\text{e-}06 = 90,162$ (turbulent, $Re > 4,000$)

Friction: $h_f = 10.67 L Q^{1.852} / (C^{1.852} D^{4.8704}) = 10.67 \times 494.0 \times 0.00011 / (7,090.0 \times 0.00001) = 5.94 \text{ m}$

Minor losses: $h_m = k_m h_f = 0.10 \times 5.94 = 0.59 \text{ m}$; total $h_L = 5.94 + 0.59 = 6.53 \text{ m}$ (gradient 13.22 m/km)

HGL: $H_{\text{start}} = 1687.25 \text{ m}$, $H_{\text{end}} = 1687.25 - 6.53 = 1680.72 \text{ m}$; pressure along the pipe 58.4 to 58.4 m

Surge (Joukowsky): $\Delta h = aV/g = 1000 \times 0.917 / 9.81 = 93.5 \text{ m}$; minimum transient pressure $\approx p_{\text{min}} - \Delta h = 58.4 - 93.5 = -35.1 \text{ m}$

Pressure class (working pressure): PN10 (0-494 m); ER: PN10 494 m

Pipe CHAPWA_BRANCH: J-004N → CHAPWA_GSR04 (DN100, L = 2,790 m, C = 120)

Design flow: $Q = V_d / (t \times 3600) = 622.10 / (24 \times 3600) = \mathbf{0.0072 \text{ m}^3/\text{s}}$ (7.2 l/s)

Area: $A = \pi D^2/4 = \pi \times 0.100^2 / 4 = 0.0079 \text{ m}^2$; velocity $V = Q/A = 0.0072 / 0.0079 = \mathbf{0.917 \text{ m/s}}$

Reynolds number: $Re = VD/\nu = 0.917 \times 0.100 / 1.017\text{e-}06 = 90,162$ (turbulent, $Re > 4,000$)

Friction: $h_f = 10.67 L Q^{1.852} / (C^{1.852} D^{4.8704}) = 10.67 \times 2,790.5 \times 0.00011 / (7,090.0 \times 0.00001) = \mathbf{33.53 \text{ m}}$

Minor losses: $h_m = k_m h_f = 0.10 \times 33.53 = 3.35 \text{ m}$; total $h_L = 33.53 + 3.35 = \mathbf{36.88 \text{ m}}$ (gradient 13.22 m/km)

HGL: $H_{\text{start}} = 1687.25 \text{ m}$, $H_{\text{end}} = 1687.25 - 36.88 = 1650.37 \text{ m}$; pressure along the pipe 58.4 to 127.8 m
Surge (Joukowsky): $\Delta h = aV/g = 1000 \times 0.917 / 9.81 = \mathbf{93.5 \text{ m}}$; minimum transient pressure $\approx p_{\text{min}} - \Delta h = 58.4 - 93.5 = -35.1 \text{ m}$

Pressure class (working pressure): PN10 (0-599 m), PN16 (599-2,587 m), PN25 (2,587-2,790 m); ER: PN25 2,664 m

Controlling point: pipe UHURU_J004N at 179 m (pipe crown 1698.32 m); the HGL there must be $\geq 1698.32 + 2.0 = 1700.32 \text{ m}$, so at the start of UHURU_J004N: $H = 1700.32 + 10.35 \times 179/1,563 = 1701.51 \text{ m}$

Head required at the start of the zone: $H_{\text{req}} = \mathbf{1701.51 \text{ m}}$

Available head: Uhuru Park BWL = 1697.60 m; surplus = $\mathbf{-3.9 \text{ m}}$

The negative surplus does not mean the tanks cannot be reached. Chapwa arrives with 125 m of excess head and Makambini with 21 m. The shortfall is local, in the first ~200 m after the GBR: the surveyed ground ($\approx 1,698\text{--}1,699 \text{ m}$) lies above the GBR bottom water level taken from the ER elevation (1,697.3 m). The pressure at the pipe crown there falls below the 2 m criterion. The outlet pipe should be laid deeper over this length, or the GBR levels confirmed.

7. Hydraulic Profiles

Figures 2 and 3 show the longitudinal sections: the surveyed ground and the computed HGL. At each pump station the HGL rises vertically by the pump head. At each tank it drops to the water level. Wherever the HGL stays above the ground line the pressure is positive.

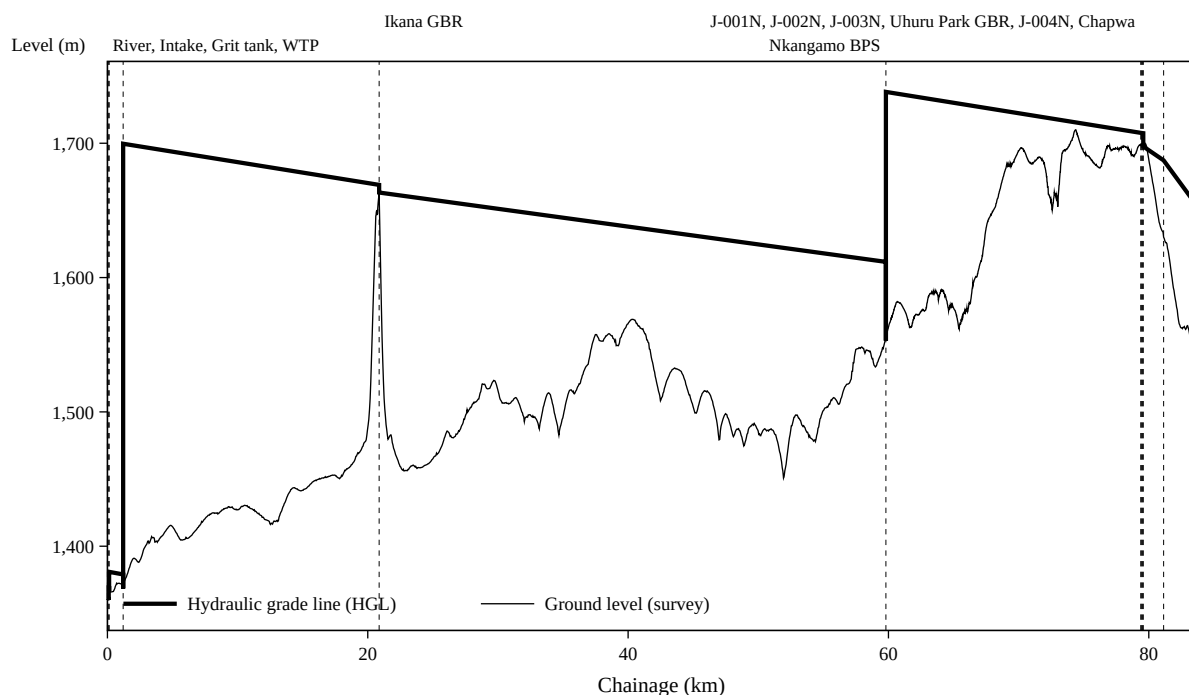


Figure 2. Longitudinal profile from the Momba River to Chapwa GSR 04 (via the WTP, Ikana GBR, Nkangamo BPS and Uhuru Park GBR).

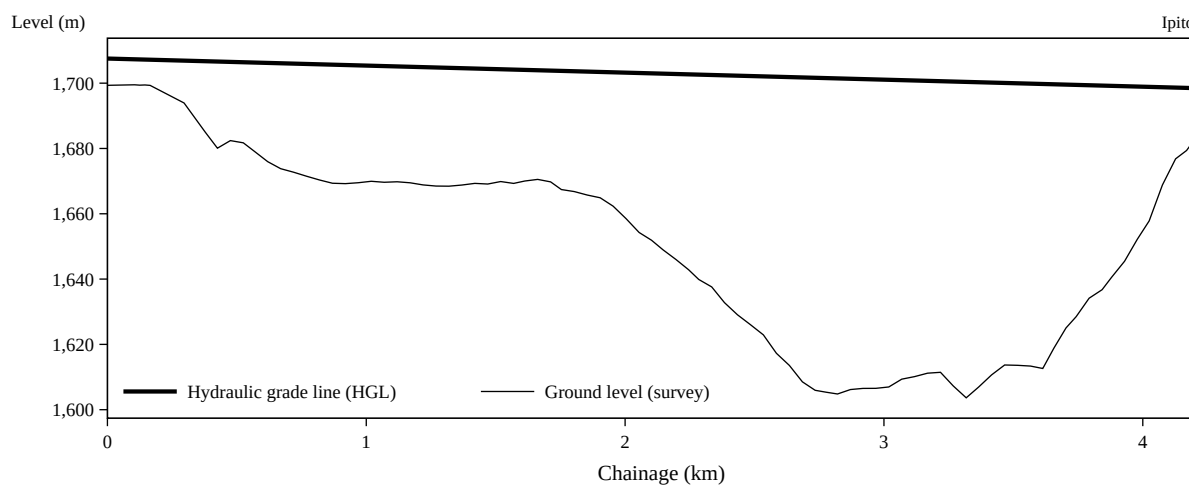


Figure 3. Longitudinal profile of the Ipito branch (J-003N to Ipito GSR 02).

A high point of the BPS main at about 74.1 km (ground $\approx 1,710$ m) lies above the Uhuru Park top water level. The HGL must stay above it during pumping, and air valves are required. When the pumps stop, the main will partly drain towards both ends.

8. Pressure Classes, Surge and Storage

Table 7. Pipe length (m) per pressure class: ER schedule vs this analysis (working pressure), and the unprotected Joukowsky surge.

Pipe	ER schedule	This analysis	Δh (m)
ABSTRACTION	PN10: 100	PN10: 100	98
RAW_MAIN	PN10: 1,177	PN10: 1,078	98
WTP_IKANA_RM	PN10: 305, PN16: 119, PN25: 4,705, PN30: 10,798, PN40: 5,067	PN10: 257, PN16: 436, PN25: 6,746, PN30: 10,776, PN40: 1,450	91
IKANA_NKANGAMO_GM	PN10: 2,877, PN16: 13,247, PN25: 22,380	PN10: 570, PN16: 18,575, PN25: 19,784	83
BPS_RM_1	PN10: 10,310, PN16: 7,177, PN25: 4,185	PN10: 12,023, PN16: 5,624, PN25: 1,935	91
ESR_BRANCH	PN10: 38	PN10: 38	61
GSR01_BRANCH	PN16: 40	PN10: 40	113
IPITO_BRANCH	PN10: 4,285	PN10: 3,986, PN16: 234	45
UHURU_J004N	PN10: 1,555	PN10: 1,563	83
MAKAMBINI_BRANC H	PN10: 494	PN10: 494	93
CHAPWA_BRANCH	PN25: 2,664	PN10: 599, PN16: 1,988, PN25: 204	93

The PN30 length of the WTP → Ikana main agrees closely with the ER. The ER's PN40 length corresponds to the length that needs PN40 once surge is included. Unprotected surge would reach 80–100 m on the long mains and produce sub-atmospheric transients. Surge vessels at the WTP and BPS pump stations and air valves at the high points are therefore required, as the ER specifies. Their sizing requires a transient analysis.

Table 8. Balancing storage, Equation (12).

Tank	Inflow (m ³ /d)	t (h)	V_{bal} (m ³)	Volume (m ³)	Storage (h)
IKANA_GBR	20,000	22	$20,000 \times (1 - 22/24) = 1,667$	3,000	3.6
UHURU_PARK_GBR	17,460	22	$17,460 \times (1 - 22/24) = 1,455$	5,000	6.9
TUNDUMA_ESR	372	22	$372 \times (1 - 22/24) = 31$	120	7.8
UHURU_GSR01	1,547	22	$1,547 \times (1 - 22/24) = 129$	500	7.8
IPITO_GSR02	622	22	$622 \times (1 - 22/24) = 52$	200	7.7
MAKAMBINI_GSR03	622	24	$622 \times (1 - 24/24) = 0$	200	7.7
CHAPWA_GSR04	622	24	$622 \times (1 - 24/24) = 0$	200	7.7

Every tank has more than the balancing volume required by the 22-hour pumping regime. The Nkangamo sump (2,000 m³) receives 24-hour gravity flow and pumps for 22 hours; it must hold $20,000 \times 2/24 = 1,667$ m³ over the two pump-off hours, which it can.

9. Verification with EPANET

The network was rebuilt as an EPANET 2.2 model with the same surveyed profile. It includes pumps, tanks at their critical levels, and flow-control valves at the tank inlets. It was solved twice: with pump curves through the duty points computed in Section 6, and with pump curves through the ER duty points.

Table 9. Pump operating points from EPANET compared with the design.

Pump curves	Station	EPANET Q (m ³ /h)	Design Q (m ³ /h)	EPANET head (m)	Design TDH (m)
Computed duty	MOMBA_INTAKE	1178	978	10.2	12.1
Computed duty	GRIT_TANK	980	978	17.4	17.4
Computed duty	MOMBA_WTP	913	909	333.3	334.1
Computed duty	NKANGAMO_BPS	909	909	188.2	188.2
ER duty	MOMBA_INTAKE	1065	978	9.4	12.1
ER duty	GRIT_TANK	1230	978	20.1	17.4
ER duty	MOMBA_WTP	902	909	332.5	334.1
ER duty	NKANGAMO_BPS	909	909	221.1	188.2

Table 10. HGL at the end of selected pipes: EPANET (computed pump duties) vs the Python analysis.

Pipe	EPANET (m)	Python (m)	Difference (m)
RAW_MAIN	1379.00	1379.00	+0.00
BPS_RM_1	1707.78	1707.73	+0.06
BPS_RM_2	1707.72	1707.67	+0.06
BPS_RM_3	1707.59	1707.54	+0.06
ESR_BRANCH	1707.56	1707.50	+0.06
GSR01_BRANCH	1707.26	1707.20	+0.06
IPITO_BRANCH	1698.49	1698.44	+0.05

With the computed pump duties, EPANET reproduces the design flows and the HGL at the junctions of the BPS main within about 0.1 m. This confirms the hand-traceable calculations. The intake low-lift pump shows a larger flow in EPANET, because the grit tank is held at its mid level there. With the ER pump curves, the WTP pumps deliver within 1 % of the design flow, which confirms the ER selection. The grit-tank and BPS pumps have more head than required. Their flow must be controlled, by the inlet valves or by variable-speed drives.

10. Review of the Draft EPANET Model

The draft model produced from the route drawing was checked against the survey and the ER. Its ground levels agree with the survey to within 0.01 m. It is not, however, a valid hydraulic model of the system, for the following reasons:

- the source reservoir head equals the ground level (1,365.94 m) instead of the river water level;
- no pumps are defined;
- the WTP, Ikana GBR, Nkangamo BPS and three existing tanks are missing;
- node N2163 is modelled as a tank although it is junction J-003N;
- all pipes are placeholders (300 mm, C = 120) instead of DN600 steel with DN150/DN100 branches;
- tank geometry is a placeholder (D = 15 m, 0–6 m);
- there are no demands and no flow-control valves;
- two inferred connectors (up to 34 % slope) replace unsurveyed sections;

- two pipes are shorter than the distance between their end nodes;
- the WTP–Ikana and BPS–Uhuru Park sections are 1.3–2.1 km shorter than the ER lengths.

Solved as it stands, the draft returns pressures down to –209 m and reverse flow towards the river. The corrected model described in this report resolves these deficiencies.

11. Conclusions and Recommendations

- The system delivers 20,000 m³/d to the Tunduma tanks with a raw-water abstraction of 21,505 m³/d. All velocities (0.4–1.1 m/s) are within the ER limit of 3 m/s.
- The WTP clear-water pumps require TDH = 334 m at 303 m³/h per pump (3 duty). This agrees with the ER duty of 330 m; each pump needs a 450 kW motor. The station uses 24,575 kWh/d (1.23 kWh/m³).
- The Nkangamo BPS requires TDH = 188 m, governed by the existing ESR. The ER value of 221 m has about 33 m of reserve. The ESR levels should be confirmed before the pumps are purchased.
- The Ikana–Nkangamo gravity main has 53 m of surplus head. A flow-control valve is required at the BPS sump inlet.
- The branches to Ipito, Makambini and Chapwa arrive with excess heads of 13, 21 and 125 m. They require flow-control or pressure-reducing valves; the lower Chapwa section requires PN25.
- Surge protection is required on all long mains, together with air valves at the high points (notably ≈74 km on the BPS main). A transient analysis should size the surge vessels.
- The assumed operating levels (river, intake, grit tank, WTP, tanks) and the missing route sections should be confirmed. The analysis can then be re-run directly from the configuration file.

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